JASON LU

Research engineer: verified agentic data → RL environments & benchmarks (3D, game dev, AI research) → self-improving AI

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EDUCATION

Columbia University | *M.S. in Computer Science* — *New York, NY* | Expected Dec. 2027

NYU Shanghai | *B.S. in Computer Science* | *GPA 3.84 / 4.00* — *Shanghai, China* | Jun. 2026

EXPERIENCE

Evolvent AI — May 2026 – Present

Research Assistant, Recursive Self-Improvement Infrastructure — *Shanghai, China*

- **Agentic data synthesis engine**: Architected the production, benchmark-agnostic engine behind Evolvent's recurring data deliveries to **ByteDance, Alibaba (Qwen), and Zhipu AI (GLM)**: multi-agent orchestration turns benchmarks and real-world engineering issues into agentic post-training data, and closed-loop verification checks every trajectory before it ships.
- **3D asset-repair benchmark** (bought by 3 top AI companies): **1,200+ tasks** in which agents use the Blender API to repair meshes carrying 14 injected production defects (3 coupling tiers) into game-engine-ready assets that meet a poly/quad/UV/pivot/collider spec. Grading is fully programmatic, no VLM: 31 checks (manifold, UV overlap, texel density, headless Godot import) plus a Hausdorff bound against the original mesh that blocks delete-to-pass hacks; **96%** agreement with human review. The best of 6 frontier models fixes **72%** of single defects but only **24%** of 3-coupled ones; 41% of failures break UVs after retopology.
- **3D level-reconstruction RL sandbox** (bought by 3 top AI companies): Agents rebuild a beatable Godot 3D platformer from 1–6 screenshots; frontier models reach only **34%** beatable on medium. Path-first generator (reachable route, then platforms, obstacles, occlusion; 8–60 objects) guarantees 100% solvable levels and infinite curricula. Reward = Hungarian-matched layout score + navmesh-bot playability, exposing **38%** of rebuilds scoring >0.8 on layout as unbeatable; closed 5 reward-hacking classes (floor carpeting, goal beside spawn) via an isolated verifier and hidden checks. CPU-only, ∼40 s/rollout, 64 parallel.
- **Agentic GameDev benchmark** (bought by top AI companies): From only a gameplay spec, coding agents must ship a complete, runnable Godot 4 3D game; **12 tasks, 724 criteria** over Mechanics, Depth, Visuals, and Art (25% each); 96 builds by 4 frontier models average only ∼**48/100**, weakest on Mechanics (∼30): they look right but don't play right. An agent-written adapter binds the rubric to each project's native runtime state, so fixed tests reproducibly verify raycast hits, collisions, and resource settlement; VLM judgments of recordings are cross-checked against that state. Rubrics are calibrated on 12 hidden reference games.
- **RSIGym** [paper · code]: Co-led a budgeted environment in which frontier agents run AI research end to end (data synthesis→LoRA training→eval); at $500/benchmark the best agent took a 35B base model's **SWE-bench Verified 17.67% → 50.33%** and AIME 31.67% → 97.78%. Co-designed **RSI-Index** (mean share of remaining gap closed across 5 benchmarks; best agent **0.4809**).
- **RSIBench-Data** [paper · code]: Co-designed a frozen-budget benchmark casting agents as post-training data researchers (35B MoE, **4 agents × 6 benchmarks**): **SWE-bench Verified 12% → 46%**, yet 18 of 23 searches that ran past their peak ended lower.
- **Gödel Forest** [paper · code]: Co-built a multi-agent RSI system in which parallel search trees co-evolve by exchanging ∼100-token lessons: **+10.70 pp** over a single agent across 6 benchmarks (**SWE-bench Verified 35% → 54%**).

Intent (GroupUltra) — Feb. 2026 – Aug. 2026

Agent Systems Engineering Intern, personal & workplace agents — *Remote*

- **Agentic 3D ad generation** (commercialized): From a product's 3D model (or one generated from a photo) and an ad brief, an agent autonomously composes scene, lighting, materials, and camera moves in Blender and renders a hero image plus a 15-second ad video in 16:9/9:16/1:1. Designed its **23 tools** for 3D perception and action (preview renders as observations, spatial measurement, scene diff, checkpoint/rollback), 12 lighting and 8 camera-move skills, and a generate→check→fix loop in which programmatic checks (framing, clipping, logo safe zone, exposure, brand color) plus VLM review trigger self-rework.
- **Personal & team agents**: Scaled to **10K+ users** with **80%+** autonomous decisions; shipped a shared multiplayer team agent that any group-chat member can @-mention, with channel-wide memory, tools, async long tasks, and proactive follow-ups.
- **Durable agent runtime**: **99%+** failure recovery across 100K+ jobs and 1M+ tool calls, with <0.01% duplicate critical effects, on **24 h+** tasks: designed the restart-safe runtime (checkpoints, leases with watchdogs, idempotency keys, side-effect fencing) and gated releases on deterministic replay of production traces (core-task success **+30%**, manual intervention −60%).

DP Technology — Mar. 2025 – Feb. 2026

Algorithm Engineering Intern, Uni-Parser team — *Shanghai, China*

- **Uni-Parser** [paper · site]: Core contributor to the industrial multi-expert engine that turns papers and patents into LLM and AI4Science training corpora; built its pipeline-parallel inference path (micro-batching, async GPU dispatch, load balancing) and scaled it to **240 L40 GPUs**, parsing **16M+ documents in 6 days** with near-linear scaling to 1,000 GPUs.
- **MolParser-Mobile** [paper · model]: Co-designed a **9.98M-param** optical chemical structure recognition (OCSR) model that runs **38× faster** than MolParser 1.0 at higher accuracy (1,520 mol/s on one GPU; cached-K/V cross-attention decoder) and ran its full training recipe: pretraining on 18M+ image→E-SMILES pairs, SFT, on-policy distillation, and DPO targeting chirality and Markush errors (pattern-match **52% → 78%**).

RESEARCH PROJECTS

CrashGym | *Python, NumPy, scikit-learn* — NeurIPS 2026 Agenthon Workshop · [paper]

- Built a deterministic order-book crash simulator (price impact, liquidity withdrawal, liquidations) where LLM agents experiment on a lab copy before intervening, scored by regret over every feasible action; lab access cut regret **57.4%** on 24 worlds.

SKILLS

Training: PyTorch, TRL, LoRA/QLoRA SFT, DPO | **Infra:** Python, Rust, C/C++, TypeScript, Blender/Godot, Harbor, E2B, Docker